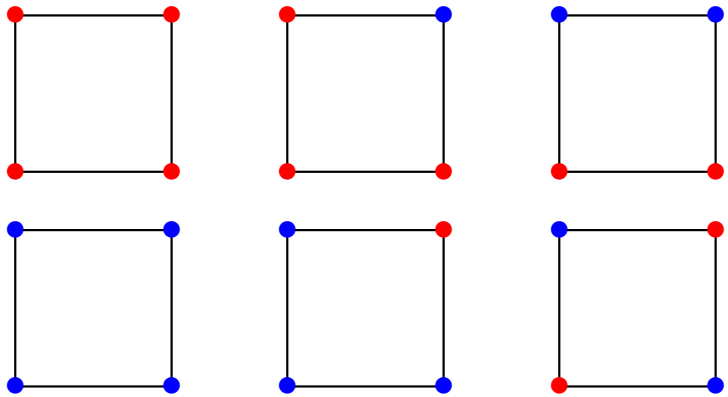
Introduction

- Colorings of a regular tetrahedron.
- Colorings of a square.



Coloring in the presence of symmetries.



Permutation and symmetry groups I

- Let X be a finite set. Without loss of generality, we take X to be the set

  $$\{1, 2, \ldots, n\},$$

  consisting of the first n positive integers. Each permutation

  $$i_1, i_2, \ldots, i_n$$

  of X can be viewed as a one-to-one function from X to itself defined by

  $$f : X \to X$$

  where

  $$f(1) = i_1, f(2) = i_2, \ldots, f(n) = i_n.$$

  By the pigeonhole principle, each one-to-one function $f : X \to X$ is onto. We also denote this permutation by the 2-by-n array

  $$\begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{pmatrix}.$$



Permutation and symmetry groups II

- We denote the set of all $n!$ permutations of $\{1, 2, \ldots, n\}$ by S_n. Since permutations are now functions, they can be combined, using composition, that is, following one by another. If

$$f = \begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{pmatrix}.$$

and

$$g = \begin{pmatrix} 1 & 2 & \cdots & n \\ j_1 & j_2 & \cdots & j_n \end{pmatrix}.$$

are two permutations of $\{1, 2, \ldots, n\}$, then their *composition*, in the order f followed by g, is the permutation

$$g \circ f = \begin{pmatrix} 1 & 2 & \cdots & n \\ j_1 & j_2 & \cdots & j_n \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{pmatrix},$$



Permutation and symmetry groups III

where

$$g \circ f(k) = g(f(k)) = j_{i_k}.$$

Composition of functions defines a *binary operation* on S_n: *If f and g are in S_n, then $g \circ f$ is also in S_n.*

- The binary operation $\circ$ of composition of permutations in S_n satisfies the *associative law*

$$(f \circ g) \circ h = f \circ (g \circ h),$$

but it does not satisfy the commutative law. In general,

$$f \circ g \neq g \circ f,$$

although equality may hold in some instances. We use the usual power notation to denote compositions of a permutation with itself,

$$f^1 = f, f^2 = f \circ f, f^3 = f \circ f \circ f, \cdots, f^k = \underbrace{f \circ f \circ \cdots \circ f}_{k}.$$



Permutation and symmetry groups IV

The *identity permutation* is the permutation ι of $\{1, 2, \ldots, n\}$ that takes each integer to itself,

$$\iota(k) = k \text{ for all } k = 1, 2, \ldots, n;$$

equivalently,

$$\iota = \begin{pmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{pmatrix}.$$

Obviously, $\iota \circ f = f \circ \iota = f$ for all permutations f in S_n.

- Each permutation in S_n, since it is a one-to-one function, has an inverse f^{-1} that is also a permutation in S_n,

$$f^{-1}(k) = s, \text{ provided that } f(s) = k.$$

The 2-by-n array for f^{-1} can be gotten from the 2-by-n array for f by interchanging rows 1 and 2 and then rearrange columns so that the integers $1, 2, \ldots, n$ occur in the natural order in the first row.



Permutation and symmetry groups V

- For each permutation f we define $f^0 = \iota$. The inverse of the identity permutation is itself: $\iota^{-1} = \iota$. The definition of inverse implies that, for all f in S_n, we have $f \circ f^{-1} = f^{-1} \circ f = \iota$.
- A *group of permutations of* X, for short a *permutation group*, is defined to be a nonempty subset G of permutations in S_n satisfying the following three properties.
  1. (*closure under composition*) For all permutations f and g in G, $f \circ g$ is also in G.
  2. (*identity*) The identity permutation ι of S_n belongs to G.
  3. (*closure under inverses*) For each permutation f in G the inverse f^{-1} is also in G.
- The set S_n of all permutations of $X = \{1, 2, \ldots, n\}$ is a permutation group, called the *symmetric group of order* n.
- At the other extreme, the set $G = \{\iota\}$ consisting only of the identity permutation is a permutation group.



Permutation and symmetry groups VI

- Every permutation group satisfies the *cancelation law*

  $f \circ g = f \circ h$ implies that $g = h$.

- There is an abstract concept of a group, which is defined to be a nonempty set with a binary operation, which satisfies the association law and also (i) closure under composition, (ii) identity, and (iii) closure under inverse. Permutation groups are groups.

Example

Let n be a positive integer, and let ρ_n denote the permutation of $\{1, 2, \ldots, n\}$ defined by

$$\rho_n = \begin{pmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 2 & 3 & 4 & \cdots & n & 1 \end{pmatrix}.$$



Permutation and symmetry groups VII

Thus, $\rho_n(i) = i + 1$ for $i = 1, 2, \ldots, n-1$, and $\rho_n(n) = 1$. This implies

$$\rho_n^k = \begin{pmatrix} 1 & 2 & \cdots & n-k & n-k+1 & \cdots & n \\ k+1 & k+2 & \cdots & n & 1 & \cdots & k-1 \end{pmatrix}$$

for each nonnegative integer k. In particular, if r equals $k \mod n$, then $\rho_n^r = \rho_n^k$. Thus, there are only n distinct powers of ρ_n, namely,

$$\rho_n^0 = \iota, \rho_n, \rho_n^2, \ldots, \rho_n^{n-1}.$$

We thus conclude that

$$C_n = \{\rho^0 = \iota, \rho_n, \rho_n^2, \ldots, \rho_n^{n-1}\}$$

is a permutation group. It is an example of a *cyclic group* of order n.

- Let Ω be a geometrical figure. A *symmetry* of Ω is a (geometric) motion or congruence that brings the figure Ω onto itself.
  - The geometric figures are composed of corners (or vertices) and edges, and in the case of three-dimensional figures, of faces (or sides).



Permutation and symmetry groups VIII

- Each symmetry acts as a permutation on the corners, on the edges, and, in the three-dimensional figures, on the faces.
- A symmetry of Ω followed by another, that is, the composition of two symmetries, is again a symmetry.
- Similarly, the inverse of a symmetry is also a symmetry.
- Finally, the motion that leaves everything fixed is a symmetry, the identity symmetry.

Hence, we conclude that the symmetries of Ω act as a permutation group G_C on its corners, a permutation group G_E on its edges, and, in the case where Ω is three-dimensional, a permutation group G_F on its faces.



Permutation and symmetry groups IX

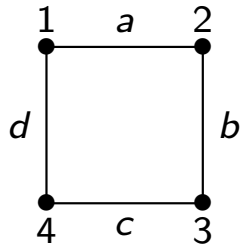
- As a result, a set of permutations that results by considering all the symmetries of a figure is automatically a permutation group. Thus, we have a *corner-symmetry group*, an *edge-symmetry group*, a *face-symmetry group*, and so on.

Example

Consider a square Ω with its corners labeled 1, 2, 3, and 4 and its edges labeled a, b, c, and d, as in Figure. There are 8 symmetries of Ω and there are of two types. There are the 4 rotations about the center of the square through the angles of 0, 90, 180, and 270 degrees. These 4 symmetries constitute the *planar symmetries* of Ω, the symmetries where the motion takes place in the plane containing Ω. The planar symmetries by themselves form a group. The other symmetries are the four reflections about the lines joining opposite corners and the lines joining the midpoints of opposite sides. For these symmetries the motion takes place in space since to "flip" the square one needs to go outside of the plane containing it.



Permutation and symmetry groups X



- The rotations acting on the corners give the four permutations

$$\rho_4^0 = \iota = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \rho_4 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

$$\rho_4^2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \rho_4^3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}.$$

The reflection acting on the corners give the four permutations

$$\tau_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \tau_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

$$\tau_3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \tau_4 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}.$$



Permutation and symmetry groups XI

Thus, the corner-symmetry group of a square is

$$G_C = \{\rho_4^0 = \iota, \rho_4, \rho_4^2, \rho_4^3, \tau_1, \tau_2, \tau_3, \tau_4\}.$$

We check that $\tau_3 = \rho_4 \circ \tau_1$, $\tau_2 = \rho_4^2 \circ \tau_1$, and $\tau_4 = \rho_4^3 \circ \tau_1$. Hence we can also write

$$G_C = \{\rho_4^0 = \iota, \rho_4, \rho_4^2, \rho_4^3, \tau_1, \rho_4 \circ \tau_1, \rho_4^2 \circ \tau_1, \rho_4^3 \circ \tau_1\}.$$

- In a similar way we can obtain the symmetry group of a regular n-gon for any $n \geq 3$. Besides, the n rotations $\rho_n^0 = \iota, \rho_n, \ldots, \rho_n^{n-1}$, we have n reflections $\tau_1, \tau_2, \ldots, \tau_n$. If n is even, then there are $n/2$ reflections about opposite corners and $n/2$ reflections about the lines joining the midpoints of opposite sides. If n is odd, then the reflections are the n reflections about the lines joining a corner to the side opposite it. The resulting group

$$D_n = \{\rho_n^0 = \iota, \rho_n, \ldots, \rho_n^{n-1}, \tau_1, \tau_2, \ldots, \tau_n\}$$

of $2n$ permutations of $\{1, 2, \ldots, n\}$ is an instance of a *dihedral group* of order $2n$.



Permutation and symmetry groups XII

- Suppose we have a group G of permutations of a set $X = \{1, 2, \ldots, n\}$ of the first n positive integers. A *coloring* of X is an assignment of a color to each element of X. Let $\mathcal{C}$ be a collection of colorings of X.
- The set $\mathcal{C}$ can be *any* collection of colorings of X as long as G takes a coloring in $\mathcal{C}$ to another coloring in $\mathcal{C}$ in the manner to be described.
- Let $\boldsymbol{c}$ be a coloring of X in which the colors of $1, 2, \ldots, n$ are denoted by $c(1), c(2), \ldots, c(n)$, respectively. Let

$$f \begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{pmatrix}$$

be a permutation in G. Then $f * \boldsymbol{c}$ is defined to be the coloring in which i_k has the color $c(k)$; that is,

$$(f * c)(i_k) = c(k),$$



Permutation and symmetry groups XIII

or using the inverse of f,

$$(f * c)(k) = c(f^{-1}(k)).$$

In words, since g moves k to i_k, the color of k, namely $c(k)$, moves to $f(k) = i_k$ and becomes the color of i_k. The set $\mathcal{C}$ of colorings is required to have the property that, for all f in G and all $\boldsymbol{c}$ in $\mathcal{C}$, $f * \boldsymbol{c}$ is also in $\mathcal{C}$. This implies that f permutes the colorings in $\mathcal{C}$, and thus denotes the coloring in $\mathcal{C}$ into which $\boldsymbol{c}$ is sent by f.

- The basic relationship that holds between the two operations $\circ$ (composition of permutations in G) and $*$ (action of permutations in G on colorings in $\mathcal{C}$) is

$$(g \circ f) * \boldsymbol{c} = g * (f * \boldsymbol{c}).$$

The left side of equation is the coloring in which the color of k moves to $(g \circ f)(k)$. The right side is the coloring in which the color of k moves to $f(k)$ and then moves to $g(f(k))$.



Permutation and symmetry groups XIV

- Let $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$ be two colorings in $\mathcal{C}$. We define a relation called *equivalence*, denoted by $\sim$ on $\mathcal{C}$ as follows: $\boldsymbol{c}_1$ is *equivalent* (under action of G) to $\boldsymbol{c}_2$, provided there is a permutation f in G such that

$$f * \boldsymbol{c}_1 = \boldsymbol{c}_2$$

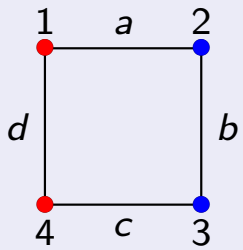
Two colorings are *inequivalent*, provided that they are not equivalent.



Permutation and symmetry groups XV

Example

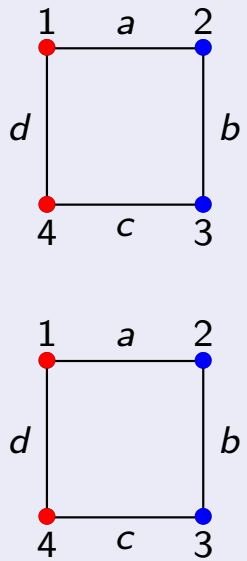
Permutation in G_c	Effect on the coloring (R, B, B, R)
$\rho_4^0 = \iota$	(R, B, B, R)
ρ_4	(R, R, B, B)
ρ_4^2	(B, R, R, B)
ρ_4^3	(B, B, R, R)
τ_1	(R, R, B, B)
τ_2	(B, B, R, R)
τ_3	(B, R, R, B)
τ_4	(R, B, B, R)



Permutation and symmetry groups XV

Example

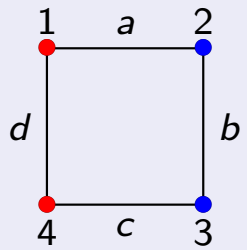
Permutation in G_c	Effect on the coloring (R, B, B, R)	
$\rho_4^0 = \iota$	(R, B, B, R)	$\leftarrow$
ρ_4	(R, R, B, B)	
ρ_4^2	(B, R, R, B)	
ρ_4^3	(B, B, R, R)	
τ_1	(R, R, B, B)	
τ_2	(B, B, R, R)	
τ_3	(B, R, R, B)	
τ_4	(R, B, B, R)	

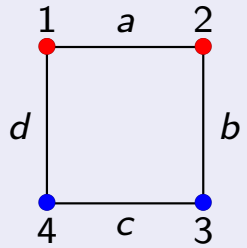


Permutation and symmetry groups XV

Example

Permutation in G_c	Effect on the coloring (R, B, B, R)	
$\rho_4^0 = \iota$	(R, B, B, R)	
ρ_4	(R, R, B, B)	$\leftarrow$
ρ_4^2	(B, R, R, B)	
ρ_4^3	(B, B, R, R)	
τ_1	(R, R, B, B)	
τ_2	(B, B, R, R)	
τ_3	(B, R, R, B)	
τ_4	(R, B, B, R)	

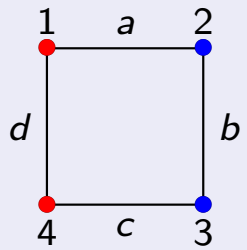


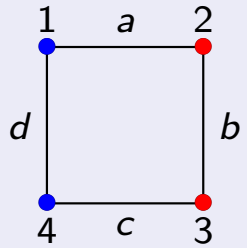


Permutation and symmetry groups XV

Example

Permutation in G_c	Effect on the coloring (R, B, B, R)	
$\rho_4^0 = \iota$	(R, B, B, R)	
ρ_4	(R, R, B, B)	
ρ_4^2	(B, R, R, B)	$\leftarrow$
ρ_4^3	(B, B, R, R)	
τ_1	(R, R, B, B)	
τ_2	(B, B, R, R)	
τ_3	(B, R, R, B)	
τ_4	(R, B, B, R)	

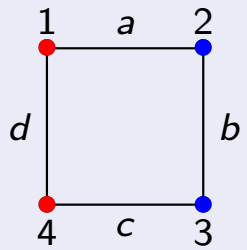


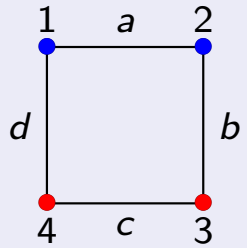


Permutation and symmetry groups XV

Example

Permutation in G_c	Effect on the coloring (R, B, B, R)
$\rho_4^0 = \iota$	(R, B, B, R)
ρ_4	(R, R, B, B)
ρ_4^2	(B, R, R, B)
ρ_4^3	(B, B, R, R) $\leftarrow$
τ_1	(R, R, B, B)
τ_2	(B, B, R, R)
τ_3	(B, R, R, B)
τ_4	(R, B, B, R)

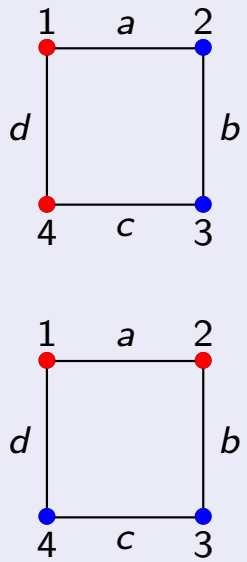




Permutation and symmetry groups XV

Example

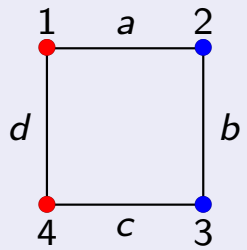
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$\rho_4^0 = \iota$	(R, B, B, R)	
ρ_4	(R, R, B, B)	
ρ_4^2	(B, R, R, B)	
ρ_4^3	(B, B, R, R)	
τ_1	(R, R, B, B)	$\leftarrow$
τ_2	(B, B, R, R)	
τ_3	(B, R, R, B)	
τ_4	(R, B, B, R)	

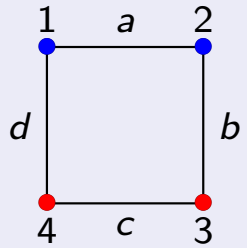


Permutation and symmetry groups XV

Example

Permutation in G_c	Effect on the coloring (R, B, B, R)	
$\rho_4^0 = \iota$	(R, B, B, R)	
ρ_4	(R, R, B, B)	
ρ_4^2	(B, R, R, B)	
ρ_4^3	(B, B, R, R)	
τ_1	(R, R, B, B)	
τ_2	(B, B, R, R)	$\leftarrow$
τ_3	(B, R, R, B)	
τ_4	(R, B, B, R)	

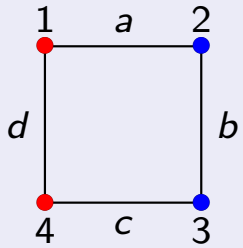


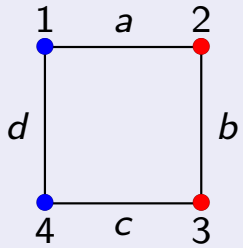


Permutation and symmetry groups XV

Example

Permutation in G_c	Effect on the coloring (R, B, B, R)	
$\rho_4^0 = \iota$	(R, B, B, R)	
ρ_4	(R, R, B, B)	
ρ_4^2	(B, R, R, B)	
ρ_4^3	(B, B, R, R)	
τ_1	(R, R, B, B)	
τ_2	(B, B, R, R)	
τ_3	(B, R, R, B)	$\leftarrow$
τ_4	(R, B, B, R)	

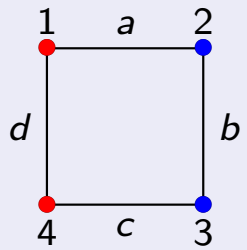


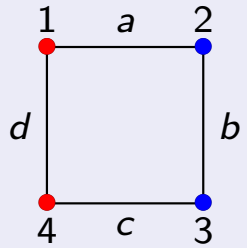


Permutation and symmetry groups XV

Example

Permutation in G_c	Effect on the coloring (R, B, B, R)	
$\rho_4^0 = \iota$	(R, B, B, R)	
ρ_4	(R, R, B, B)	
ρ_4^2	(B, R, R, B)	
ρ_4^3	(B, B, R, R)	
τ_1	(R, R, B, B)	
τ_2	(B, B, R, R)	
τ_3	(B, R, R, B)	
τ_4	(R, B, B, R)	$\leftarrow$





Burnside's theorem I

- We derive and apply a formula of (Burnside, 1911, p.191) for counting the number of inequivalent colorings of a set X under the action of a group of permutation of X.
- It is possible, that for an appropriate choice of f and $\boldsymbol{c}$, we have

$$f * \boldsymbol{c} = \boldsymbol{c}.$$

If we allow either f to vary over all permutations in G or $\boldsymbol{c}$ to vary over all colorings in $\mathcal{C}$, then we get both

$$G(\boldsymbol{c}) = \{f : f \in G, f * \boldsymbol{c} = \boldsymbol{c}\},$$

the set of all permutations in G that fix the coloring $\boldsymbol{c}$, and

$$\mathcal{C}(f) = \{\boldsymbol{c} : \boldsymbol{c} \in \mathcal{C}, f * \boldsymbol{c} = \boldsymbol{c}\},$$



Burnside's theorem II

the set of all colorings in $\mathcal{C}$ that are fixed by f. The set $G(\boldsymbol{c})$ of all permutations that fix the coloring $\boldsymbol{c}$ is called the *stabilizer* of $\boldsymbol{c}$. The stabilizer of any coloring also forms a group of permutations.

Theorem

For each coloring $\boldsymbol{c}$, the stabilizer $G(\boldsymbol{c})$ of $\boldsymbol{c}$ is a permutation group. Moreover, for any permutations of f and g in G, $g * \boldsymbol{c} = f * \boldsymbol{c}$ if and only if $f^{-1} \circ g$ is in $G(\boldsymbol{c})$.



Burnside's theorem III

Corollary

Let $\boldsymbol{c}$ be a coloring in $\mathcal{C}$. The number

$$|\{f * \boldsymbol{c} : f \in G\}|$$

of different colorings that are equivalent to $\boldsymbol{c}$ equals the number

$$\frac{|G|}{|G(\boldsymbol{c})|}$$

obtained by dividing the number of permutations in G by the number of permutations in the stabilizer of $\boldsymbol{c}$.



Burnside's theorem IV

Theorem

Burnside's Theorem – Let G be a group of permutations of X and let C be a set of colorings of X such that $f * \boldsymbol{c}$ is in $\mathcal{C}$ for all f in G and all $\boldsymbol{c}$ in $\mathcal{C}$. Then the number $N(G, \mathcal{C})$ of nonequivalent colorings in $\mathcal{C}$ is given by

$$N(G, \mathcal{C}) = \frac{1}{|G|} \sum_{f \in G} |\mathcal{C}(f)|.$$

In words, the number of nonequivalent colorings in $\mathcal{C}$ equals the average of the number of colorings fixed by the permutations in G.

Example

Counting circular permutations – How many ways are there to arrange n distinct objects in a circle?

The answer is the number of ways to color the corners of a regular n-gon Ω with n different colors that are nonequivalent with respect to the group of



Burnside's theorem V

rotations of Ω. Let $\mathcal{C}$ consist of all $n!$ ways to color the n corners of Ω in which each of the n colors occurs once. Then the cyclic group

$$C_n = \{\rho_n^0 = \iota, \rho_n, \cdots, \rho_n^{n-1}\}$$

acts on $\mathcal{C}$, and the number of circular permutations equals the number of nonequivalent colorings in $\mathcal{C}$. The identity permutation ι in C_n fixes all $n!$ of the colorings in $\mathcal{C}$. Every other permutation in $\mathcal{C}$ does not fix any coloring in $\mathcal{C}$. Hence

$$N(C_n, \mathcal{C}) = \frac{1}{n}(n! + 0 + \cdots + 0) = (n-1)!.$$

Example

Counting necklaces – How many ways are there to arrange $n \geq 3$ differently colored beads in a necklace?



Burnside's theorem VI

We have almost the same situation as described in the previous example, except since necklaces can be flipped over, the group G of permutation now has to be taken to be the entire vertex-symmetry group of a regular n-gon. Thus, G is the dihedral group D_n of order $2n$. The only permutation that can fix a coloring is the identity and it fixes all $n!$ colorings. Hence, the number of nonequivalent colorings is

$$N(D_n, \mathcal{C}) = \frac{1}{2n}(n! + 0 + \cdots + 0) = \frac{(n-1)!}{2}.$$

Example

How many nonequivalent ways are there to color the corners of a regular 5-gon with the colors red and blue?

The group of symmetries of a regular 5-gon is the dihedral group

$$D_5 = \{\rho_5^0 = \iota, \rho_5, \rho_5^2, \rho_5^3, \rho_5^4, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5\}.$$



Burnside's theorem VII

We have

$$|\mathcal{C}(\rho_5^i)| = \begin{cases} 2^5 = 32, & i = 0, \\ 2, & i = 1, 2, 3, 4, \end{cases}$$

$$|\mathcal{C}(\tau_j)| = 2 \times 2 \times 2 = 8, \quad j = 1, 2, 3, 4, 5.$$

Hence

$$N(D_5, \mathcal{C}) = \frac{1}{10}(2^5 + 2 + 2 + 2 + 2 + 8 + 8 + 8 + 8 + 8) = 8.$$



Pólya's counting formula I

- As we have seen in the previous section, success in using Burnside's theorem for counting the number of nonequivalent colorings in the presence of a permutation group G acting on a set $\mathcal{C}$ of colorings is dependent on being able to compute the number $|\mathcal{C}(f)|$ of colorings in $\mathcal{C}$ fixed by a permutation f in G. This computation can be facilitated by consideration of the cyclic structure of a permutation.
- Let f be a permutation of $X = \{1, 2, \cdots, n\}$. Let $D_f = (X, A_f)$ be the digraph whose set of vertices is X and whose set of arcs is $A_f = \{(i, f(i)) : i \in X\}$. The digraph has n vertices and n arcs. Moreover, the indegree and outdegree of each vertex equal 1. The set A_f of arcs can be partitioned into directed cycles, with each vertex belonging exactly one directed cycle.



Pólya's counting formula II

Example

Let

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 6 & 8 & 5 & 4 & 1 & 3 & 2 & 7 \end{pmatrix}$$

be a permutation of $\{1, 2, \cdots, 8\}$. We obtained the following partition of D_f into directed cycles, $1 \to 6 \to 3 \to 5 \to 1$, $2 \to 8 \to 7 \to 2$, $4 \to 4$. Let us write

$$[1\ 6\ 3\ 5] = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 6 & 2 & 5 & 4 & 1 & 3 & 7 & 8 \end{pmatrix}.$$

We call such a permutation, in which certain of the elements are permuted in a cycle and the remaining elements, if any, are fixed, a *cycle permutation* or, more briefly, a *cycle*. If the number of elements in the cycle is k, then we call it a *k-cycle*. We now observe that the partition of D_f into directed cycles



Pólya's counting formula III

corresponds to a factorization (with respect to the composition $\circ$) of f into permutation cycles, $f = [1\ 6\ 3\ 5] \circ [2\ 8\ 7] \circ [4]$.

- Let f be any permutation of the set X. Then, we see that, with respect to the operation of composition, f has a factorization

$$f = [i_1\ i_2\ \cdots\ i_p] \circ [j_1\ j_2\ \cdots\ j_q] \circ \cdots \circ [l_1\ l_2\ \cdots\ l_r] \tag{14.1}$$

into cycles, where each integer in X occurs in exactly one of the cycles. We call (14.1) the *cycle factorization* of f. The cycle factorization of f is unique, apart from the order in which the cycles appear, and this order is arbitrary. In the cycle factorization of a permutation of X, every element of X occurs exactly once.

Example

Cycle factorization of each permutation in the dihedral group D_4 of order 8.

Pólya's counting formula IV

D_4	Cycle factorization
$\rho_4^0 = \iota$	$[1] \circ [2] \circ [3] \circ [4]$
ρ_4	$[1\ 2\ 3\ 4]$
ρ_4^2	$[1\ 3] \circ [2\ 4]$
ρ_4^3	$[1\ 4\ 3\ 2]$
τ_1	$[1] \circ [2\ 4] \circ [3]$
τ_2	$[1\ 3] \circ [2] \circ [4]$
τ_3	$[1\ 2] \circ [3\ 4]$
τ_4	$[1\ 4] \circ [2\ 3]$

We denote by $\#(f)$ the *number of cycles in the cycle factorization* of a permutation f.



Pólya's counting formula V

Theorem

Let f be a permutation of a set X. Suppose we have k colors available with which to color the elements of X. Let $\mathcal{C}$ be the set of all colorings of X. Then the number $|\mathcal{C}(f)|$ of colorings of $\mathcal{C}$ that are fixed by f equals $k^{\#(f)}$.



Pólya's counting formula VI

Example

f in D_4	Cycle factorization	$\#(f)$	$\lvert\mathcal{C}(f)\rvert$
$\rho_4^0 = \iota$	$[1] \circ [2] \circ [3] \circ [4]$	4	$3^4 = 81$
ρ_4	$[1\ 2\ 3\ 4]$	1	$3^1 = 3$
ρ_4^2	$[1\ 3] \circ [2\ 4]$	2	$3^2 = 9$
ρ_4^3	$[1\ 4\ 3\ 2]$	1	$3^1 = 3$
τ_1	$[1] \circ [2\ 4] \circ [3]$	3	$3^3 = 27$
τ_2	$[1\ 3] \circ [2] \circ [4]$	3	$3^3 = 27$
τ_3	$[1\ 2] \circ [3\ 4]$	2	$3^2 = 9$
τ_4	$[1\ 4] \circ [2\ 3]$	2	$3^2 = 9$

The number of nonequivalent colorings is $N(D_4, \mathcal{C}) = \frac{81+3+9+3+27+27+9+9}{8} = 21$.



Pólya's counting formula VII

- Let f be a permutation of X where X has n elements. Suppose that the cycle factorization of f has e_1 1-cycles, e_2 2-cycles, $\cdots$, and e_n n-cycles. The numbers $e_1, e_2, \cdots, e_n$ are nonnegative integers satisfying

$$1e_1 + 2e_2 + \cdots + ne_n = n. \tag{14.2}$$

We call the n-tuple $(e_1, e_2, \cdots, e_n)$ the *type* of the permutation f and write

$$\text{type}(f) = (e_1, e_2, \cdots, e_n).$$

Note

$$\#(f) = e_1 + e_2 + \cdots + e_n.$$

We introduce n indeterminates

$$z_1, z_2, \cdots, z_n,$$



Pólya's counting formula VIII

where z_k is to correspond to a k-cycle ($k = 1, 2, \cdots, n$). To each permutation f with type$(g) = (e_1, e_2, \cdots, e_n)$, we associate the monomial of f,

$$\text{mon}(f) = z_1^{e_1} z_2^{e_2} \cdots z_n^{e_n}.$$

Let G be a group of permutations of X. Summing these monomials for each f in G, we get the generating function

$$\sum_{f \in G} \text{mon}(f) = \sum_{f \in G} z_1^{e_1} z_2^{e_2} \cdots z_n^{e_n} \tag{14.3}$$

for the permutations in G according to type.



Pólya's counting formula IX

- If we combine like terms in (14.3), the coefficient of $z_1^{e_1} z_2^{e_2} \cdots z_n^{e_n}$ equals the number of permutations in G of type $(e_1, e_2, \cdots, e_n)$. The *cycle index* of G

$$P_G(z_1, z_2, \cdots, z_n) = \frac{1}{|G|} \sum_{f \in G} z_1^{e_1} z_2^{e_2} \cdots z_n^{e_n}$$

  is this generating function divided by the number $|G|$ of permutations in G.



Pólya's counting formula X

Example

D_4	Cycle factorization	Type	Monomial
$\rho_4^0 = \iota$	$[1] \circ [2] \circ [3] \circ [4]$	$(4,0,0,0)$	$z_1^4 z_2^0 z_3^0 z_4^0 = z_1^4$
ρ_4	$[1\ 2\ 3\ 4]$	$(0,0,0,1)$	$z_1^0 z_2^0 z_3^0 z_4^1 = z_4$
ρ_4^2	$[1\ 3] \circ [2\ 4]$	$(0,2,0,0)$	$z_1^0 z_2^2 z_3^0 z_4^0 = z_2^2$
ρ_4^3	$[1\ 4\ 3\ 2]$	$(0,0,0,1)$	$z_1^0 z_2^0 z_3^0 z_4^1 = z_4$
τ_1	$[1] \circ [2\ 4] \circ [3]$	$(2,1,0,0)$	$z_1^2 z_2^1 z_3^0 z_4^0 = z_1^2 z_2$
τ_2	$[1\ 3] \circ [2] \circ [4]$	$(2,1,0,0)$	$z_1^2 z_2^1 z_3^0 z_4^0 = z_1^2 z_2$
τ_3	$[1\ 2] \circ [3\ 4]$	$(0,2,0,0)$	$z_1^0 z_2^2 z_3^0 z_4^0 = z_2^2$
τ_4	$[1\ 4] \circ [2\ 3]$	$(0,2,0,0)$	$z_1^0 z_2^2 z_3^0 z_4^0 = z_2^2$

The cycle index of D_4 is $P_{D_4}(z_1, z_2, z_3, z_4) = \frac{1}{8}(z_1^4 + 2z_4 + 3z_2^2 + 2z_1^2 z_2)$.



Pólya's counting formula XI

Theorem

Let X be a set of n elements, and suppose we have a set of k colors available with which to color the elements of X. Let $\mathcal{C}$ be the set of all k^n colorings of X. Let G be a group of permutations of X. Then the number of nonequivalent colorings is the number

$$N(G, \mathcal{C}) = P_G(k, k, \cdots, k)$$

obtained by substituting $z_i = k$, $(i = 1, 2, \cdots, n)$ into the cycle index of G.

Example

We are given a set of k colors. What is the number of nonequivalent ways to color the corners of a square?

The cycle index of the dihedral group D_4 is

$$P_{D_4}(z_1, z_2, z_3, z_4) = \frac{1}{8}(z_1^4 + 2z_4 + 3z_2^2 + 2z_1^2 z_2).$$



Pólya's counting formula XII

Hence, the number of nonequivalent colorings is

$$P_{D_4}(k,k,k,k) = \frac{k^4 + 2k^3 + 3k^2 + 2k}{8}.$$

If $k = 6$, $P_{D_4}(6,6,6,6) = 231$.

Theorem

Pólya's theorem – Let X be a set of n elements and let G be a group of permutation of X. Let $\{u_1, u_2, \cdots, u_k\}$ be a set of k colors and let $\mathcal{C}$ be any set of colorings of X with the property that G acts as a permutation group on $\mathcal{C}$. Then the generating function for the number of nonequivalent colorings of $\mathcal{C}$ according to the number of colors of each kind is the expression

$$P_G(u_1 + \cdots + u_k, u_1^2 + \cdots + u_k^2, \cdots, u_1^n + \cdots + u_k^n), \tag{14.4}$$



Pólya's counting formula XIII

obtained from the cycle index $P_G(z_1, z_2, \cdots, z_n)$ by making the substitutions

$$z_j = u_1^j + \cdots + u_k^j \quad (j = 1, 2, \cdots, n).$$

In other words, the coefficient of

$$u_1^{p_1} u_2^{p_2} \cdots u_k^{p_k}$$

in (14.4) equals the number of nonequivalent colorings in $\mathcal{C}$ with p_1 elements of X colored u_1, p_2 elements colored u_2, $\cdots$, p_k elements colored u_k.

Example

Determine the generating function for the number of nonequivalent colorings of the corners of a square with 2 colors and also those with 3 colors.

The cycle index of D_4 is

$$P_{D_4}(z_1, z_2, z_3, z_4) = \frac{1}{8}(z_1^4 + 2z_4 + 3z_2^2 + 2z_1^2 z_2).$$



Pólya's counting formula XIV

Let the two colors be r and b. Then the generating function is

$$\begin{aligned} &P_{D_4}(r+b, r^2+b^2, r^3+b^3, r^4+b^4) \\ &\quad = \tfrac{1}{8}[(r+b)^4 + 2(r^4+b^4) + 3(r^2+b^2)^2 + 2(r+b)^2(r^2+b^2)] \\ &\quad = r^4 + r^3 b + 2r^2 b^2 + r b^3 + b^4. \end{aligned}$$

The total number of nonequivalent colorings, namely, the sum of the coefficients, is 6. Now suppose that we have three colors r, b, and g. The generating function is

$$\begin{aligned} &P_{D_4}(r+b+g, r^2+b^2+g^2, r^3+b^3+g^3, r^4+b^4+g^4) \\ &\quad = \tfrac{1}{8}[(r+b+g)^4 + 2(r^4+b^4+g^4) \\ &\qquad +3(r^2+b^2+g^2)^2 + 2(r+b+g)^2(r^2+b^2+g^2)] \end{aligned}$$

The total number of nonequivalent colorings equals $P_{D_4}(3,3,3,3) = 21$.

